5.5a)

i)

From $t_0 = \frac{2}{3(1+w)H_0}$

when $w = 0$.

$$t_0 = \frac{2}{3H_0}$$

$$d_{hor}(t_0) = c\, t_0 \frac{3(1+w)}{1+3w}$$

when $w=0$

$$d(t_0) = 3c\, t_0 = \cancel{3}c \frac{2}{\cancel{3}H_0}$$

$$= \frac{2c}{H_0}$$

$$a(t) = \left(\frac{t}{t_0}\right)^{2/3+3w}$$

when $w=0$

$$a_m(t) = \sqrt{\left(\frac{t}{t_0}\right)^3}$$

$$= \sqrt{\left(\frac{3tH_0}{2}\right)^3}$$

$$d_p(t_0) = c t_0 \frac{3(1+w)}{1+3w}\left[1-\left(\frac{t_e}{t_0}\right)^{\frac{1+3w}{3+3w}}\right]$$

when $w=0$

$$d_p(t_0) = 3ct_0\left[1-\left(\frac{t_e}{t_0}\right)^{1/3}\right]$$

$$\left(\frac{t_e}{t_0}\right)^{2/3} = \frac{1}{1+z} \rightarrow a(t) = 1\left(\frac{t_e}{t_0}\right)^{2/3+\cancel{3w}}$$

$$t_0 = \frac{2}{H_0} \qquad\qquad = \frac{1}{1+z}$$

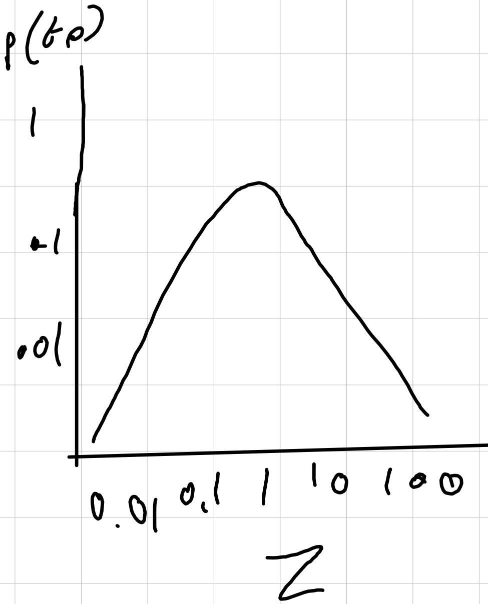
$$d_p(t_0) = 3c\left(\frac{2}{H_0}\right)\left[1-\left(\frac{1}{1+z}\right)^{1/2}\right]$$

since

$$d_p(t_e) = \frac{d_p(t_0)}{1+z}$$

$$= \frac{2c}{H_0(1+z)}\left[1-\frac{1}{\sqrt{1+z}}\right]$$

ii) $d_p(t_0)$



iii)

$$d_p(t_e) = \frac{2c}{H_0(1+3w_0)}\left[\frac{1}{1+z} - \frac{1}{3^{\frac{3(1+3w)}{2}}}\right]$$

$$\frac{d}{dz} d_p(t_e) = \frac{2c}{H_0[1+3w]}\frac{d}{dz}\left[\frac{1}{1+z} - \frac{1}{(1+z)^{\frac{3(1+w)}{2}}}\right]$$

$$= \left(\frac{3(1+w)}{2}\right)^{2/3(1+3w)} - 1$$

$w = 0$ in a matter dominated universe

$$z_{max} = \left(\frac{3}{2}\right)^2 - 1 = \frac{5}{4}$$

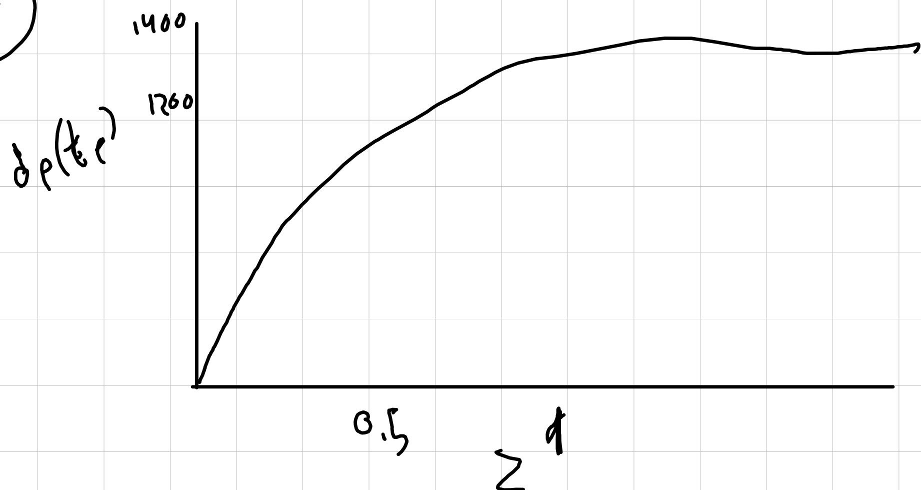
Plug into $d_p(t_e)$

$$= \frac{2c}{(1+z)H_0}\left[1 - \frac{1}{\sqrt{1+z}}\right]$$

$$= \frac{2c}{\left(1+\frac{5}{4}\right)H_0}\left[1 + - \frac{1}{\sqrt{1+\frac{5}{4}}}\right]$$

$$R_0 = \frac{8}{4}\left[1 - \frac{2}{3}\right] = 0.33$$

iv)



$$t_0 = \frac{2}{H_0(1+w) \times 3}$$

$$w = 1/3$$

$$t_0 = \frac{1}{2H_0}$$

$$d_{hor}(t_0) = c \frac{1}{\cancel{2}H_0} \frac{\cancel{4}}{\cancel{2}} = \frac{c}{H_0}$$

$$a(t) = \left(\frac{t}{t_0}\right)^{2/3+3w}$$

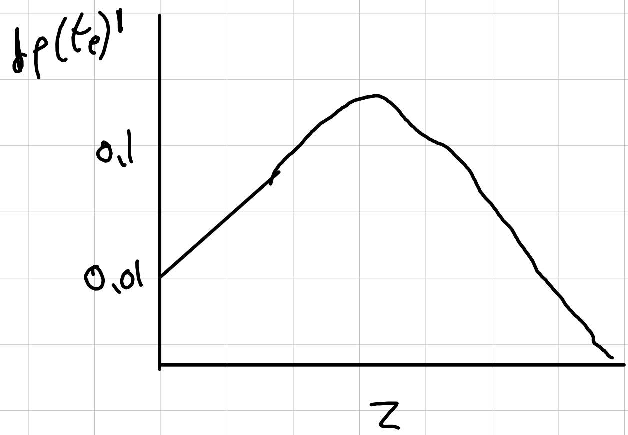
$$w = 1/3$$

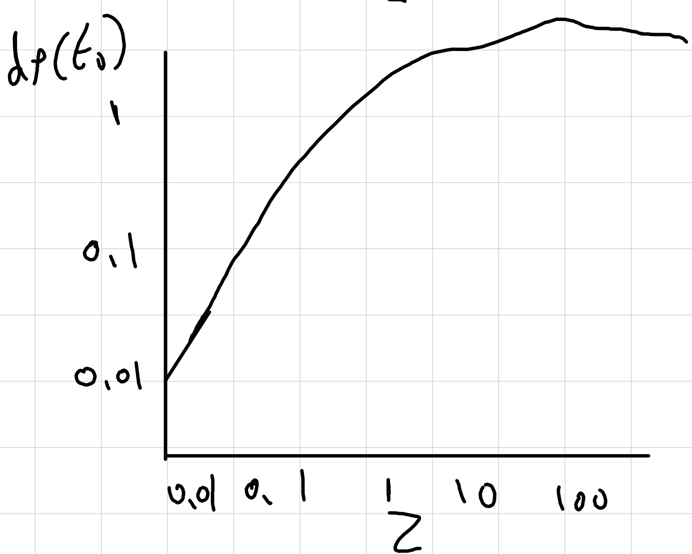
$$= \sqrt{\left(\frac{t}{t_0}\right)}$$

$$d_p(t_e) = \frac{d_p(t_0)}{(1+z)}$$

$$d_p(t_e) = \frac{c}{H_0} \frac{z}{(1+z)^2}$$

ii)





iii)

$w = 1/3$

$$z_{max} = \left[\frac{3\left(1+\frac{1}{3}\right)}{2}\right]^{2/\left(1+\frac{2}{3}\right)} - 1$$

$$= 1 \qquad \boxed{d_p(t_e) = \frac{c}{H_0}\left[\frac{z}{(1+z)^2}\right]}$$

$$= \frac{c}{H_0}\left(\frac{1}{(1+1)^2}\right) = \frac{c}{H_0}\left(\frac{1}{4}\right)$$

$$R_0 = \frac{1}{4}$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\varepsilon}{3c^2} - \frac{\kappa c^2}{R_0^2 c^2}$$

$\kappa = 0$ (Flat universe)

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\varepsilon}{3c^2} - \cancelto{0}{\frac{\kappa c^2}{R_0^2 c^2}}$$

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G\varepsilon}{3c^2}$$

$$H_0^2 = \frac{\dot{a}^2}{a^2}$$

$$\frac{\dot{a}^2}{a^2} = \boxed{\frac{8\pi G\varepsilon}{3c^2} = H_0^2}$$

given

$$\dot{a} = H_0 a$$

$$\frac{da}{a} = H dt$$

$$\int \frac{da}{a} = \int H dt$$

$$= \ln a = H(t)\Big|_{t_0}^{t}$$

$$e^{\ln a(t)} = e^{H(t)}\Big|_{t_0}^{t}$$

$$a(t) = e^{H(t-t_0)}$$

$$d_r(t_0) = \int_{t_0}^{t} \frac{dt}{a(t)}$$

$$= c\int_{t_e}^{t_0} e^{-H(t-t_0)} = c\int_{t_e}^{t} e^{H_0(t-t_0)}$$

$$= \frac{c}{H_0^2}$$

$$(1+z) = e^{H_0(t_0 - t_e)}$$

$$z = e^{H_0(t_0 - t_e)} - 1$$

$$d_p(t_e) = \frac{d_p(t_0)}{1+z}$$

$$d_p(t_e) = \frac{c}{H_0} \frac{z}{1+z}$$